

Node.js, React, and React Native Topics

Node.js Advanced Topics

1. Event Loop & Performance (phases, microtasks/macrotasks, worker_threads, child_process) 2. Asynchronous Patterns (callbacks, promises, async/await, error handling) 3. Streams & Buffers (Readable/Writable/Duplex/Transform, backpressure, Buffer) 4. Cluster & Scaling (Cluster module, PM2, load balancing) 5. Node.js Internals (V8, libuv, garbage collection, C++ addons) 6. Security (Helmet.js, preventing SQL/NoSQL injection, XSS, CSRF, rate limiting) 7. Advanced Express.js & Middleware 8. Real-time Communication (WebSockets, SSE, WebRTC) 9. Advanced APIs (GraphQL, API versioning, pagination) 10. Caching & Performance Optimization (Redis, HTTP caching, compression) 11. Testing & Debugging (Mocha, Jest, Supertest, node inspect) 12. Database Optimization (indexing, transactions, pooling) 13. Message Queues & Background Jobs (RabbitMQ, Kafka, Bull, node-cron) 14. File Uploads & Processing (multer, streaming, sharp, ffmpeg) 15. Deployment & DevOps (Docker, CI/CD, Kubernetes)

React Topics (Important for React Native)

1. JSX & Component Basics (functional vs class, props, composition) 2. State Management (useState, derived state) 3. Hooks (useState, useEffect, useRef, useContext, useReducer) 4. Props Drilling & Context API 5. Event Handling (onClick, onChange) 6. Conditional Rendering & Lists (map, keys, ternary) 7. Forms & Controlled Components 8. Component Lifecycle (via useEffect) 9. Reusability & Custom Hooks 10. Navigation Basics (concepts of routes and params)

React Native Specific Topics

1. Core Components (View, Text, ScrollView, FlatList, SectionList, Image) 2. Styling with Flexbox & StyleSheet API 3. Handling User Input (TextInput, Button, TouchableOpacity) 4. Navigation (react-navigation stack, tab, drawer) 5. Platform-specific Code (Platform API, conditional rendering) 6. Native Modules & Linking 7. Networking (fetch, axios, API calls) 8. Using Device Features (Camera, Location, Sensors) 9. State Management (Context API, Redux, Zustand) 10. Performance Optimization (memo, useCallback, useMemo, virtualization)